

Paper Photonic-Chern — Photonic Chern-Quantization of Hall Drift

Integer Quantization from Substrate $U(1)$ -Bundle Topology

Allen Proxmire

May 2026

Preamble — What This Paper Does NOT Claim

1. This paper does **not** claim derivation of integer-Chern quantization of photonic Hall drift from nothing. Result is conditional on the 13 ED primitives (Paper_087) + declared paper-specific postulates.
2. It does **not** claim closed-proof reconstruction in the Hardy 2001 / CDP 2011 / Coecke-Kissinger operational-reconstruction sense. ED sits in the substrate-ontology lineage ('t Hooft cellular-automaton, Wolfram Ruliad, causal-set program), not the operational-reconstruction lineage.
3. It does **not** override Paper_095's three-tier verdict grammar. Verdict for this paper is stated in §1.
4. It does **not** claim a new derivation of integer-Chern quantization — that follows from standard $U(1)$ -bundle topology over a closed manifold (Chern 1946) and is INHERITED. ED's substrate contribution is the **rule-type bundle identification** (P09 + T17), not the topological-quantization theorem itself.
5. **INHERITED vs FORCED breakdown:** ED supplies rule-type bundle identification (P09 $U(1)$ -valued polarity + T17 rule-type bundle, Paper_015) and Bloch-band substrate origin (Paper_011); integer quantization $C_n \in \mathbb{Z}$ is INHERITED from standard Chern-class topology; specific Chern numbers per platform are INHERITED via topological band-structure analysis.

Abstract

This paper supplies the substrate-level account of integer-Chern quantization of Hall drift in synthetic-frequency-lattice photonic systems. ED's contribution is the substrate identification of the relevant $U(1)$ bundle as a rule-type bundle (P09 + T17, Paper_015) over the closed Brillouin-zone manifold (Paper_011 Bloch theorem). The integer quantization $C_n = \int_{BZ} F_n / (2\pi) \in \mathbb{Z}$ itself follows from standard Chern-class topology (Chern 1946) and is INHERITED; the substrate framework does **not** re-derive integer-Chern quantization. Specific Chern numbers per platform / band are INHERITED via topological band-structure analysis. Per Paper_095 form-FORCED / value-INHERITED methodology, verdict is **M3** for the substrate identification of the rule-type bundle. Key falsifier **F4**: discovery of a photonic-Chern system where the (P09 + T17) rule-type-bundle identification fails to predict the observed Hall-drift quantization.

1 Statement of Result

Claim. Hall drift in synthetic-frequency-lattice photonic systems exhibits **integer Chern-quantization** as a form-IDENTIFIED consequence of:

1. Substrate-level Berry-holonomy structure (Paper_009): photonic polarity transport on rule-type bundles produces a $U(1)$ Berry connection whose integrated curvature over a Brillouin zone is quantized in units of 2π .
2. Bloch-theorem substrate translation symmetry (Paper_011): periodic substrate structure supports Bloch-band content over which the Berry curvature is integrated.
3. T17 gauge-bundle substrate origin (Paper_015): the Berry-connection $\mathcal{A}_\mu$ is a substrate-level rule-type bundle connection; its first Chern number is an integer topological invariant.

The integer-quantization $C = \int_{BZ} F/(2\pi) \in \mathbb{Z}$ of the Hall drift is form-IDENTIFIED. Specific Chern numbers per platform / band INHERITED.

Verdict: **M3**.

2 Primitive Inputs

2.1 Primitives invoked

- **P05 (Polarity-transport)** — substrate-level transport carrying photonic polarity.
- **P07 (Channel structure)** — synthetic-frequency-lattice channel content.
- **P09 ($U(1)$ -valued polarity)** — supplies the gauge group whose Chern class is quantized.
- **P10 (Rule-type primitive)** — photonic rule-type carriers.

2.2 Upstream Dependencies

- **I-009:** Berry phase as holonomy of substrate connection.
- **I-010:** Aharonov-Bohm phase from rule-type circulation.
- **I-011:** Bloch theorem from substrate translation symmetry.
- **I-015:** T17 gauge fields as rule-type bundle connections.
- **I-RQM-T7:** Lorentz representations supplying photon $(1/2, 1/2)$ vector rep.
- **I-Chern:** Standard Chern-class theory (standard math).
- **I-Synth:** Standard synthetic-frequency-lattice machinery (Yuan, Fan et al.).

3 Derivation

3.1 Synthetic-frequency lattice as substrate periodic structure

A synthetic-frequency lattice is a photonic platform where discrete frequency modes $\{\omega_n\}$ play the role of “lattice sites,” with coupling between frequency modes implemented via electro-optic modulation. The platform supports Bloch-band structure in synthetic-frequency space, with crystal-momentum-analog parameter being a phase parameter ϕ .

At substrate level, the synthetic-frequency lattice is a periodic substrate structure with periodicity in the synthetic-frequency direction. Bloch theorem (Paper_011) applies: Bloch states $\psi_n(\phi) = e^{ik\phi} u_n(\phi)$ with u_n periodic. This is the substrate-level Bloch-band content of the photonic platform.

3.2 Berry connection on photonic Bloch bands

For each Bloch band n , the **Berry connection** is defined as

$$\mathcal{A}_n(k) = i\langle u_n(k) | \partial_k u_n(k) \rangle,$$

and the **Berry curvature** as

$$F_n(k) = \partial_k \mathcal{A}_n(k) \quad (\text{scalar in 1D}),$$

or the relevant antisymmetric-tensor curvature in higher dimensions.

By Paper_009 (Berry phase as substrate-connection holonomy), $\mathcal{A}_n$ is the substrate-level rule-type-bundle connection (T17 / Paper_015) restricted to the band- n section over the parameter manifold.

3.3 Chern number as topological invariant

For a 2D Brillouin zone (real or synthetic), the **first Chern number** of band n is

$$C_n = \frac{1}{2\pi} \int_{BZ} F_n(k_x, k_y) dk_x dk_y.$$

Standard topological-quantization theorem (I-Chern): for a $U(1)$ rule-type bundle over a closed manifold (the Brillouin zone is a torus, hence closed), the integrated curvature is an **integer** in units of 2π . That is, $C_n \in \mathbb{Z}$.

ED's substrate contribution is the **identification of the $U(1)$ -bundle as a rule-type bundle** (P09 + T17), not a new derivation of integer-Chern quantization. The quantization itself follows from standard bundle topology (Chern 1946) and is INHERITED.

3.4 Hall drift from Chern numbers

In a photonic Hall-class platform, the **transverse drift response** of a wavepacket localized in band n under a force F in one synthetic-frequency direction is

$$v_{\perp} = C_n \cdot F \cdot v_{\text{unit}},$$

where v_{unit} is the band-velocity unit. The Hall drift is **quantized** in units determined by the Chern number.

The substrate-level account: the Hall drift is the substrate-Berry-curvature contribution to wavepacket motion (anomalous velocity, standard semiclassical theory). The substrate-level Berry curvature integrated over the Brillouin zone is a Chern integer; the Hall drift inherits the integer quantization.

3.5 Form-IDENTIFIED structural elements

The integer-Chern quantization is form-IDENTIFIED by:

1. **$U(1)$ rule-type bundle structure (T17 + P09).** Substrate polarity is $U(1)$ -valued; rule-type bundles are $U(1)$ -principal bundles.
2. **Closed-manifold Brillouin zone (Bloch theorem + Paper_011 substrate translation symmetry).** The Brillouin zone is topologically a torus (closed 2-manifold for 2D systems).
3. **Standard Chern-class topology (I-Chern).** Integrated curvature over closed manifold of a $U(1)$ bundle is integer-quantized.

The composition forces $C_n \in \mathbb{Z}$. The integer is form-IDENTIFIED; the specific value of C_n for a specific photonic platform's band is INHERITED via topological band-structure analysis.

3.6 Photonic-specific applications

Quantum Hall analog in synthetic frequency: Quantized transverse drift response in synthetic-frequency-lattice photonic systems (Yuan, Lin, Xiao, Fan, Minkov, etc.). Empirically realized in microresonator platforms.

Chern insulator photonics: 2D photonic crystals with broken time-reversal symmetry (gyrotropic media) exhibit edge-state propagation with Chern number $|C| \geq 1$. Substrate-level: same $U(1)$ -bundle topology.

Floquet topological photonics: Time-modulated photonic systems where the Floquet-Brillouin-zone Chern number is the topological invariant. Substrate-level: time-periodic substrate-translation symmetry supports Floquet Bloch-band structure with integer Chern numbers.

In each case, the integer Chern quantization is forced; specific values per platform are inherited.

3.7 Connection to Arc-R Berry-holonomy

Paper_009 (Berry phase) derives the Berry phase as the holonomy of the substrate polarity-transport connection around a closed loop in parameter space. This paper extends to **integrated curvature** over a closed 2-manifold:

- Single-loop holonomy $\rightarrow$ Berry phase (Paper_009).
- Integrated curvature over closed surface $\rightarrow$ Chern number (this paper).

Both inherit from the substrate $U(1)$ rule-type bundle structure (T17). The Chern integer is the global topological invariant; the Berry phase is the local holonomy. They are related by Stokes' theorem: the holonomy around a closed loop equals the integrated curvature over a surface bounded by the loop (modulo 2π).

4 Audit Table

#	Step	Label	Notes
1	P05, P07, P09, P10 primitives	P	Primitives.
2	Synthetic-frequency lattice as periodic substrate	I	Paper_011 + photonic platform.
3	Bloch band structure on synthetic lattice	I	Paper_011.
4	Berry connection $\mathcal{A}_n(k) = i\langle u \partial_k u \rangle$	I	Paper_009 + Berry theory.
5	T17 $U(1)$ rule-type bundle	I	Paper_015.
6	Berry curvature $F_n = \partial \mathcal{A}_n$	D-via-I	Standard math.
7	BZ topology = torus (closed 2-manifold)	I	Standard math.
8	Chern number $C_n = \int_{BZ} F_n / (2\pi)$	I	Standard math (I-Chern).
9	$C_n \in \mathbb{Z}$ integer quantization	D-via-I	$U(1)$ -bundle + closed manifold.
10	Hall drift $v_\perp = C_n \cdot F \cdot v_{\text{unit}}$	I	Standard semiclassical theory.
11	Quantized transverse drift response	D-via-I	ED supplies bundle ID, not quantization
12	Specific Chern numbers per platform/band	I	Empirical / band analysis.
13	Connection to Berry-phase / Aharonov-Bohm	I	Papers_009, 010.
14	Verdict M3	A $\rightarrow$ position	Per Paper_095.

5 Falsification Criteria

- **F1:** Empirical detection of non-integer Hall drift in photonic Chern-class platforms (in regimes where standard topological theory predicts integer quantization). Refutes substrate-Chern identification.

- **F2:** Substrate evidence that the photonic rule-type bundle is NOT $U(1)$ -principal. Refutes step 5.
- **F3:** Brillouin zone topology shown to be open (not torus) in photonic platforms. Refutes step 7.
- **F4:** Discovery of a photonic-Chern system where the rule-type bundle identification (P09 + T17) fails to predict the observed Hall-drift quantization correctly would falsify the ED substrate identification. (The integer-quantization itself is standard topology, not falsifiable by ED-specific experiment.)

6 Position Statement

This paper sits within the **substrate-ontology research-program lineage** ('t Hooft cellular-automaton interpretation of QM; Wolfram Ruliad / hypergraph-rewriting; causal-set program, Sorkin et al.), **not** within the operational-reconstruction lineage (Hardy 2001, CDP 2011, Coecke-Kissinger). Methodology per Paper_095 (form-FORCED / value-INHERITED).

This paper supplies the substrate-level derivation of integer-Chern quantization in photonic Hall-drift platforms, addressing the previously-flagged corpus gap.

Relationship to other papers:

- **Paper__009 (Berry phase):** parent paper supplying the substrate-Berry-holonomy structure.
- **Paper__011 (Bloch theorem):** supplies the periodic-substrate Bloch-band content.
- **Paper__015 (T17):** supplies the $U(1)$ rule-type bundle structure.

Numerical content INHERITED. Specific Chern numbers per platform. **Form IDENTIFIED.** Integer-Chern quantization of Hall drift from $U(1)$ -bundle topology (no D rows in audit table; downgraded from FORM-FORCED per Paper_095 §2.3).

Verdict: **M3.**